Oracle Clusterware OCR and OLR Relocation Procedure:

This document demonstrates the step-by-step procedure to relocate the Oracle Cluster Registry (OCR) and Oracle Local Registry (OLR) to a new ASM diskgroup in an Oracle 19c RAC environment. Proper planning and execution are critical to ensure cluster stability and avoid service disruption. The following practical session walks through the validated commands and checks required to perform the relocation safely.
Any modification or relocation activity should be thoroughly tested in a non-production environment first. It is strongly recommended to consult Oracle Support and follow official Oracle documentation before performing this procedure in a production system.

Business Case for OCR and OLR Relocation

Relocation of the Oracle Cluster Registry (OCR) and Oracle Local Registry (OLR) may be required as part of routine infrastructure maintenance, storage optimization, or risk mitigation activities in an Oracle RAC environment. Because these files are critical for cluster operation, organizations typically plan such changes during controlled maintenance windows with proper validation.

- **Storage Migration or Hardware Refresh**
  When existing ASM disks or storage arrays are being decommissioned or replaced, OCR and OLR must be moved to the new storage to maintain cluster availability and supportability.
- **Diskgroup Reorganization or Standardization**
  Organizations often consolidate or redesign ASM diskgroups (for example, moving from external to normal redundancy). Relocating OCR/OLR ensures compliance with new storage standards and improves fault tolerance.
- **Insufficient Redundancy or Risk Mitigation**
  If OCR or OLR resides on a diskgroup that does not meet redundancy best practices, relocation to a properly protected diskgroup reduces the risk of cluster failure due to disk loss.
- **Capacity or Performance Constraints**
  In environments experiencing storage pressure or performance bottlenecks, moving OCR/OLR to a healthier diskgroup helps maintain Clusterware responsiveness and stability.
- **Disaster Recovery and Compliance Requirements**
  Some organizations mandate placement of critical cluster files on dedicated or highly protected storage tiers. Relocation ensures alignment with DR policies and audit requirements.

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TASK:

Move OCR and OLR from DISKGROUP OCR(external redundancy) to OCRNORM(Normal Redundancy).

Follow the below steps to relocate OCR:

STEP 1: Check available backup of OCR

ocrconfig -showbackup

```
[root@rac01 ~]# ocrconfig -showbackup

rac01     2026/02/28 17:14:14     +OCR:/rac-cluster/OCRBACKUP/backup00.ocr.266.1226423649     724960844
```

STEP 2: Take manual backup if no recent backup is available

ocrconfig -manualbackup

```
[root@rac01 ~]# ocrconfig -manualbackup

rac02     2026/03/02 09:32:20     +OCR:/rac-cluster/OCRBACKUP/backup_20260302_093220.ocr.266.1226827941
24960844
```

STEP 3: Verify space and redundancy of ASM diskgroup

Diskgroup OCR – **External Redundancy**
Diskgroup OCRNORM – **Normal Redundancy**

```
SQL> SELECT name, state, type, total_mb, free_mb FROM v$asm_diskgroup;

NAME                 STATE       TYPE     TOTAL_MB    FREE_MB
-------------------- ----------- ------ ---------- ----------
DATA                 MOUNTED     EXTERN      20479      17417
OCRNORM              MOUNTED     NORMAL       6141       5814
OCR                  MOUNTED     EXTERN       3068       2700
RECO                 MOUNTED     EXTERN      10239       6073
```

STEP 4: Add a new OCR copy location in the specified ASM diskgroup.

```
ocrconfig -add +OCRNORM
```

```
[root@rac01 ~]# ocrconfig -add +OCRNORM
[root@rac01 ~]#
```

STEP 5: Verify the new location

```
ocrconfig
```

```
[root@rac01 ~]# ocrcheck
Status of Oracle Cluster Registry is as follows :
         Version                  :          4
         Total space (kbytes)     :     491684
         Used space (kbytes)      :      84452
         Available space (kbytes) :     407232
         ID                       : 1170265556
         Device/File Name         :       +OCR
                                    Device/File integrity check succeeded
         Device/File Name         :   +OCRNORM
                                    Device/File integrity check succeeded

                                    Device/File not configured

                                    Device/File not configured

                                    Device/File not configured

         Cluster registry integrity check succeeded

         Logical corruption check succeeded
```

STEP 6: Remove old OCR location

```
ocrconfig -delete +OCR
```

```
[root@rac01 ~]# ocrconfig -delete +OCR
[root@rac01 ~]#
```

STEP 7: Verify new location(The Old location is removed)

```
[root@rac01 ~]# ocrcheck
Status of Oracle Cluster Registry is as follows :
         Version                  :          4
         Total space (kbytes)     :     491684
         Used space (kbytes)      :      84452
         Available space (kbytes) :     407232
         ID                       : 1170265556
         Device/File Name         :    +OCRNORM
                                    Device/File integrity check succeeded

                                    Device/File not configured

                                    Device/File not configured

                                    Device/File not configured

                                    Device/File not configured

         Cluster registry integrity check succeeded

         Logical corruption check succeeded
```

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STEP 8: Take manual backup immediately

ocrconfig -manualbackup

```
[root@rac01 ~]# ocrconfig -manualbackup

rac02     2026/03/02 09:57:31     +OCRNORM:/rac-cluster/OCRBACKUP/backup_20260302_095731.ocr.258.12268
29451     724960844
```

Follow the below steps to relocate VOTING disk :

STEP 1 : Check the current location of Voting Disk:

crsctl query css votedisk

```
[root@rac01 ~]#  crsctl query css votedisk
##  STATE    File Universal Id                File Name Disk group
--  -----    -----------------                --------- ---------
 1. ONLINE   c7e555eaf7d64f14bf63b7c4cefa09d5 (/dev/oracleasm/disks/ASMDISK1) [OCR]
Located 1 voting disk(s).
```

STEP 2 : Relocate voting disk online :

crsctl replace votedisk +OCRNORM

```
[root@rac01 ~]# crsctl replace votedisk +OCRNORM
Successful addition of voting disk ac4275a331404ff2bf2bf2d7cae4b509.
Successful addition of voting disk 3d1b25a277c34fa5bf8b9c022b53ccda.
Successful addition of voting disk 6be14c91ceee4fcbbf84612632e40897.
Successful deletion of voting disk c7e555eaf7d64f14bf63b7c4cefa09d5.
Successfully replaced voting disk group with +OCRNORM.
CRS-4266: Voting file(s) successfully replaced
[root@rac01 ~]#
```

STEP 3 : Validate new location of voting disk:

```
crsctl query css votedisk
```

```
[root@rac01 ~]#  crsctl query css votedisk
##  STATE    File Universal Id                File Name Disk group
--  -----    -----------------                --------- ---------
 1. ONLINE   ac4275a331404ff2bf2bf2d7cae4b509 (/dev/oracleasm/disks/ASMDISK4) [OCRNORM]
 2. ONLINE   3d1b25a277c34fa5bf8b9c022b53ccda (/dev/oracleasm/disks/ASMDISK5) [OCRNORM]
 3. ONLINE   6be14c91ceee4fcbbf84612632e40897 (/dev/oracleasm/disks/ASMDISK6) [OCRNORM]
Located 3 voting disk(s).
```

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